PDAN 8411

POE Part 2

Plan Document

1. Justification of Dataset Suitability

The Breast Cancer Wisconsin Dataset is suitable for this study because it is a well-structured, numerical dataset designed specifically for binary classification (benign vs malignant tumours). It contains clinically relevant features such as cell size, shape, and nuclei characteristics, which are commonly used in medical diagnosis. This makes it appropriate for supervised learning, where the goal is to predict a known target variable In addition, the dataset is clean, widely used in research, and provides enough variability to effectively train and evaluate classification models.

2. Justification of Algorithm Choice (Decision Tree)

The Decision Tree algorithm was selected because it is simple, interpretable, and suitable for medical decision-support systems. It mimics human decision-making by splitting data based on feature thresholds, making it easy to explain predictions to non-technical stakeholders such as medical professionals. According to supervised learning theory, Decision Trees are effective baseline models for classification problems but may require tuning to reduce overfitting. This makes them ideal for initial modelling and comparison against more advanced ensemble methods.

3. Task Execution Plan (Pre-Analysis Plan)

The analysis will follow a structured supervised learning workflow to ensure a clear and systematic approach to model development and evaluation.

- **Data Import & Setup:** The dataset (from Kaggle) and required Python libraries will be loaded to enable analysis and modelling.
- **Exploratory Data Analysis (EDA):** The structure, data types, missing values, and distributions will be examined to understand the dataset.
- **Data Cleaning:** Data type issues will be corrected, missing or invalid values handled, and irrelevant columns removed.

- **Feature Selection:** Predictor variables (X) and target variable (y) will be defined based on medical relevance.
- **Train-Test Split:** The dataset will be split into training (80%) and testing (20%) sets to ensure unbiased evaluation.
- **Baseline Model Training:** A Decision Tree classifier will be trained as an initial benchmark model.
- **Model Optimisation:** The Decision Tree will be retrained with tuned hyperparameters to improve generalisation.
- **Benchmark Models:** Random Forest and Logistic Regression models will be trained for performance comparison.
- **Model Evaluation:** All models will be assessed using accuracy, precision, recall, F1-score, and confusion matrices.
- **Final Interpretation:** The best-performing model will be identified and interpreted in relation to the medical scheme objective.